

Answer ALL questions. Write your answers in the spaces provided.

Unless otherwise indicated, whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

1. A particle P of mass 0.5 kg is moving with velocity $(4\mathbf{i} + \mathbf{j}) \text{ m s}^{-1}$ when it receives an impulse $(2\mathbf{i} - \mathbf{j}) \text{ N s}$.

Show that the kinetic energy gained by P as a result of the impulse is 12 J .

(6)

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1. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.]

A particle A has mass 3 kg and a particle B has mass 2 kg .

The particles are moving on a smooth horizontal plane when they collide directly.

Immediately **before** the collision, the velocity of A is $(3\mathbf{i} - \mathbf{j})\text{ m s}^{-1}$ and the velocity of B is $(-6\mathbf{i} + 2\mathbf{j})\text{ m s}^{-1}$

Immediately **after** the collision the velocity of A is $\left(-2\mathbf{i} + \frac{2}{3}\mathbf{j}\right)\text{ m s}^{-1}$

- (a) Find the total kinetic energy of the two particles **before** the collision. (3)
- (b) Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the impulse exerted on A by B in the collision. (3)
- (c) Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the velocity of B immediately **after** the collision. (3)

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1. A particle P of mass 0.5 kg is moving with velocity $(4\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$ when it receives an impulse $\mathbf{J} \text{ N s}$. Immediately after receiving the impulse, P is moving with velocity $(-\mathbf{i} + 6\mathbf{j}) \text{ m s}^{-1}$.

(a) Find the magnitude of $\mathbf{J}$.

(4)

The angle between the direction of the impulse and the direction of motion of P immediately before receiving the impulse is α°

(b) Find the value of α

(3)

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1. A particle P of mass 2 kg is moving with velocity $(-4\mathbf{i} + 3\mathbf{j})\text{ m s}^{-1}$ when it receives an impulse $(-6\mathbf{i} + 42\mathbf{j})\text{ N s}$.

(a) Find the speed of P immediately after receiving the impulse.

(4)

The angle through which the direction of motion of P has been deflected by the impulse is α°

(b) Find the value of α

(2)

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3. A particle P , of mass 0.5 kg , is moving with velocity $(4\mathbf{i} + 4\mathbf{j})\text{ m s}^{-1}$ when it receives an impulse $\mathbf{I}$ of magnitude 2.5 N s .

As a result of the impulse, the direction of motion of P is deflected through an angle of 45°

Given that $\mathbf{I} = (\lambda\mathbf{i} + \mu\mathbf{j})\text{ N s}$, find all the possible pairs of values of λ and μ .

(9)

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3.

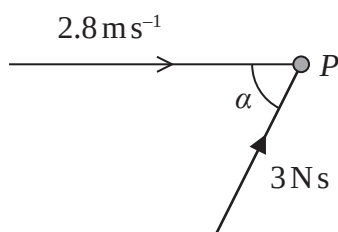


Figure 2

A particle P of mass 0.5 kg is moving in a straight line with speed 2.8 ms^{-1} when it receives an impulse of magnitude 3 Ns .

The angle between the direction of motion of P immediately before receiving the impulse and the line of action of the impulse is α , where $\tan \alpha = \frac{4}{3}$, as shown in Figure 2.

Find the speed of P immediately after receiving the impulse.

(5)

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4. A particle P has mass 0.5 kg . It is moving in the xy plane with velocity $8\mathbf{i} \text{ m s}^{-1}$ when it receives an impulse $\lambda(-\mathbf{i} + \mathbf{j}) \text{ N s}$, where λ is a positive constant.

The angle between the direction of motion of P immediately before receiving the impulse and the direction of motion of P immediately after receiving the impulse is θ°

Immediately after receiving the impulse, P is moving with speed $4\sqrt{10} \text{ m s}^{-1}$

Find (i) the value of λ

(ii) the value of θ

(8)

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6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors in a horizontal plane.]

A smooth uniform sphere A has mass 0.2 kg and another smooth uniform sphere B , with the same radius as A , has mass 0.4 kg .

The spheres are moving on a smooth horizontal surface when they collide obliquely. Immediately before the collision, the velocity of A is $(3\mathbf{i} + 2\mathbf{j})\text{ m s}^{-1}$ and the velocity of B is $(-4\mathbf{i} - \mathbf{j})\text{ m s}^{-1}$

At the instant of collision, the line joining the centres of the spheres is parallel to $\mathbf{i}$

The coefficient of restitution between the spheres is $\frac{3}{7}$

(a) Find the velocity of A immediately after the collision.

(7)

(b) Find the magnitude of the impulse received by A in the collision.

(2)

(c) Find, to the nearest degree, the size of the angle through which the direction of motion of A is deflected as a result of the collision.

(3)



4.

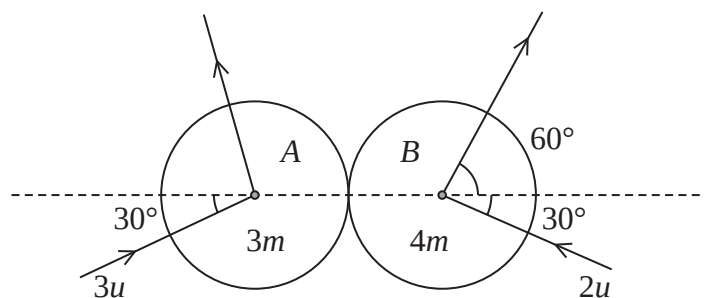


Figure 3

Two smooth uniform spheres, A and B , have equal radii. The mass of A is $3m$ and the mass of B is $4m$. The spheres are moving on a smooth horizontal plane when they collide obliquely. Immediately before they collide, A is moving with speed $3u$ at 30° to the line of centres of the spheres and B is moving with speed $2u$ at 30° to the line of centres of the spheres. The direction of motion of B is turned through an angle of 90° by the collision, as shown in Figure 3.

- Find the size of the angle through which the direction of motion of A is turned as a result of the collision.
- Find, in terms of m and u , the magnitude of the impulse received by B in the collision.

(9)

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